Assignments for the Dynamical Systems Course

July 1, 2026

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1 Assignments set IV

1.1

(Elliptical trapping region for the Lorenz system) This exercise considers an alternative bounding function given by

$$L_3(x, y, z) = \frac{rx^2 + \sigma y^2 + \sigma(z - 2r)^2}{2} \tag{1}$$

a Calculate $\dot{L_3}$.

b Find a value $C_3 > 0$ such that, if $L_3(x, y, z) > C_3$, then $\dot{L_3} < 0$.

c Conclude that all solutions enter and remain in the region

$$U_3 = \{(x, y, z) : L_3(x, y, z) \leq C_3\} \tag{2}$$

(i.e., that U_3 is a trapping region).

1.2

Consider the system of differential equations given in polar coordinates by

$$\begin{aligned} \dot{r} =& r(1 - r^2)(r^2 - 4) \\ \dot{\theta} =& 1. \end{aligned} \tag{3}$$

a Does the system have sensitive dependence on initial conditions at points in either of the sets $\{r = 1\}$ or $\{r = 2\}$. Explain your answer.

b Does the system have sensitive dependence on initial conditions when restricted to either of the sets $\{r = 1\}$ or $\{r = 2\}$. Explain your answer.

c Is either of the sets $\{r = 1\}$ or $\{r = 2\}$ a chaotic attractor?

1.3

Rikitake made a model to explain the reversal of the magnetic poles of the earth. The system of differential equations is

$$\dot{x} = -\mu x + yz \tag{4}$$

$$\dot{y} = -\mu y + (z - A)x \tag{5}$$

$$\dot{z} = 1 - xy. \tag{6}$$

a Show that there are two fixed points, where x* satisfies

$$(x^*)^2 - (x^*)^{-2} = \frac{A}{\mu}, \tag{7}$$

$$y^* = \frac{1}{x^*}, \quad \text{and} \tag{8}$$

$$z^* = \mu(x^*)^2. \tag{9}$$

b Show that the characteristic equation of the linearization at the fixed points is

$$\lambda^3 + 2\mu\lambda^2 + \frac{A}{\mu}\lambda + 2A = 0. \tag{10}$$

Show that $\lambda_1 = -2\mu$ is one eigenvalue, and the other two eigenvalues are $\pm i\sqrt{\left(\frac{A}{\mu}\right)}$.

c Show that the total system decreases the volume as t increases.